Lecture 17

Antiferromagnet

$$\vec{n} = \lambda_i \vec{N}(x_i,\tau)\sqrt{1-\left(\frac{a^d}{S}\right)^2 \vec{L}^2(x_i,\tau)} + \left(\frac{a^d}{S}\right)\vec{L}(x_i,\tau)$$

here $\vec{N}(x,\tau)$ = <u>staggered</u> magnetization (Néel vector)

$\vec{L}(x,\tau)$ = <u>uniform</u> magnetization

(normalization explained later)

$$\lambda_i = \begin{cases} +1 & \text{on A sublattice} \\ -1 & \text{on B} \end{cases}$$

$\vec{N}\cdot\vec{L} = 0$ so that $\vec{n}^2 = 1 = \vec{N}^2$

Thus, $\vec{N}$ and $\vec{L}$ are <u>smooth</u> fields that vary on distances $\gg$ lattice spacing a [that is hoped so].

Then $\vec{n}(x,\tau,u) = \lambda_i \vec{N}(x,\tau,u) + \left(\frac{a^d}{S}\right)\vec{L}(x,\tau,u) + O\left(\frac{a^d}{S}\right)^2$

so that $\vec{N}(x,\tau,u=1) \equiv \vec{N}(x,\tau)$ = physical field

$\vec{N}(x,\tau,u=0)$ = const (say, $(0,0,1)$) = gauge choice

<u>$d=1$</u>

$\lambda_i = (-1)^{x_i}$

$A_B = iS\int d\tau \int du \left(\vec{n}\cdot\partial_u\vec{n}\times\partial_\tau\vec{n}\right)$ on every site

So, $A_B = A_B(x)$.

$$\Rightarrow \text{total } A = iS\sum_x \Big\{(-1)^x \int d\tau\int du\, \vec{N}\cdot\partial_u\vec{N}\times\partial_\tau\vec{N}$$

$$+ \int d\tau\int du\left[\left(\frac{a^d}{S}\right)\right]\Big[\vec{L}\cdot\partial_u\vec{N}\times\partial_\tau\vec{N} + \vec{N}\cdot\partial_u\vec{L}\times\partial_\tau\vec{N}$$

$$+ \vec{N}\cdot\partial_u\vec{N}\times\partial_\tau\vec{L}\Big] + \underbrace{\int d\tau\, du\left(\frac{a^d}{S}\right)^2(L^2 N)}_{\text{terms with } \vec{L}^2 \text{ and } \vec{L}^3}\Big\}$$

$\Downarrow$

<u>neglect</u>

$= A_N + A_L$

Consider $A_N = iS \sum_x (-1)^x \int d\tau\, du\; \vec{N}\cdot\partial_u \vec{N}\times\partial_\tau \vec{N} =$

$$= iS \int d\tau \int du \Big\{ \sum_{\substack{x=2r \\ (\text{even})}} \vec{N}(2r)\cdot\partial_u \vec{N}(2r)\times\partial_\tau \vec{N}(2r)$$

$$- \sum_{\substack{x=2r-1 \\ (\text{odd})}} \vec{N}(2r-1)\cdot\partial_u \vec{N}(2r-1)\times\partial_\tau \vec{N}(2r-1)\Big\}$$

$$\vec{N}(2r-1) \approx \vec{N}(2r) - a\,\partial_x \vec{N}(2r) + O(a^2).$$

Observe that $\vec{N}^2 = 1 \Rightarrow \underbrace{\partial_x \vec{N}\cdot\partial_u \vec{N}\times\partial_\tau \vec{N}}_{} = 0$

$\left.\begin{array}{l} \partial_u \vec{N}^2 = 0,\ \partial_\tau \vec{N}^2 = 0,\ \partial_x \vec{N}^2 = 0 \\ \vec{N}\cdot\partial_u\vec{N} = 0,\ \vec{N}\cdot\partial_\tau\vec{N} = 0,\ \vec{N}\cdot\partial_x\vec{N} = 0 \end{array}\right\}$ there are only 2 independent direction along which vector $\vec{N}$ can change.

$\Rightarrow \partial_x\vec{N}, \partial_u\vec{N}, \partial_\tau\vec{N}$ live in <u>one plane!</u>

$$\{\ \} = \sum_{r=1}^{N/2} \vec{N}(2r)\cdot\partial_u\vec{N}(2r)\times\partial_\tau\vec{N}(2r) -$$

$$-\left[\vec{N}(2r) - a\,\partial_x\vec{N}(2r)\right]\cdot\partial_u\left[\vec{N}(2r) - a\,\partial_x\vec{N}(2r)\right]\times$$

$$\times\,\partial_\tau\left[\vec{N}(2r) - a\,\partial_x\vec{N}(2r)\right]$$

$$= \sum_{r=1}^{N/2} a\Big\{\underbrace{\partial_x\vec{N}\cdot\partial_u\vec{N}\times\partial_\tau\vec{N}}_{\emptyset} + \vec{N}(2r)\cdot\partial_u\partial_x\vec{N}(2r)\times\partial_\tau\vec{N}(2r)$$

$$+ \vec{N}(2r)\cdot\partial_u\vec{N}(2r)\times\partial_\tau\partial_x\vec{N}(2r)\Big\}$$

$$+ O(a^2) \qquad \sum_{r=1}^{N/2} a f(2r) = \frac{1}{2}(2a)\sum_{n=2}^{N} f(n) = \frac{1}{2}\int dx\, f(x)$$

$$= \frac{1}{2}\int dx\; \vec{N}(x,\tau,u)\cdot\partial_u\partial_x\vec{N}(x,\tau,u)\times\partial_\tau\vec{N}(x,\tau,u)$$

$$+ \vec{N}(x,\tau,u)\cdot\partial_u\vec{N}(x,\tau,u)\times\partial_\tau\partial_x\vec{N}(x,\tau,u) =$$

$$= \frac{1}{2}\int dx\; \partial_u\left(\vec{N}\cdot\partial_x\vec{N}\times\partial_\tau\vec{N}\right) - \vec{N}\cdot\partial_x\vec{N}\times\partial_u\partial_\tau\vec{N}$$

$$+ \partial_\tau\left(\vec{N}\cdot\partial_u\vec{N}\times\partial_x\vec{N}\right) - \vec{N}\cdot\partial_\tau\partial_u\vec{N}\times\partial_x\vec{N}$$

add up to <u>zero</u>

$$\Rightarrow A_N = iS\,\frac{1}{2}\int dx\int_0^\beta d\tau\int_0^1 du\;\partial_u\left(\vec{N}\cdot\partial_x\vec{N}\times\partial_\tau\vec{N}\right)$$

$$+ \partial_\tau\left(\vec{N}\cdot\partial_u\vec{N}\times\partial_x\vec{N}\right)$$

But $\int dx \int du \int_0^\beta d\tau \, \partial_\tau \left(\vec{N} \cdot \partial_u \vec{N} \times \partial_x \vec{N} \right) =$

$$= \int dx \int du \left(\vec{N} \cdot \partial_u \vec{N} \times \partial_x \vec{N} \right) \Big|_0^\beta = 0$$

due to periodic BC : $\vec{N}(\tau=0) = \vec{N}(\tau=\beta)$.

Next, $\int dx \int d\tau \int_0^1 du \, \partial_u \left(\vec{N} \cdot \partial_x \vec{N} \times \partial_\tau \vec{N} \right) =$

$$= \int dx \int d\tau \; \vec{N}(x,\tau,u=1) \cdot \partial_x \vec{N}(x,\tau,u=1) \times \partial_\tau \vec{N}(x,\tau,u=1)$$

$$- \underbrace{\vec{N}(x,\tau,u=0) \cdot \partial_x \vec{N}(x,\tau,u=0) \times \partial_\tau \vec{N}(x,\tau,u=0)}_{\emptyset \text{ because } \vec{N}(x,\tau,u=0) = \text{constant!}}$$

$$= \int dx \int d\tau \; \vec{N}(x,\tau) \cdot \partial_x \vec{N}(x,\tau) \times \partial_\tau \vec{N}(x,\tau)$$

$$\Rightarrow A_N = i \frac{\theta}{4\pi} \int dx \int_0^\beta d\tau \left(\vec{N} \cdot \partial_x \vec{N} \times \partial_\tau \vec{N} \right)$$

Where $\theta = 2\pi S$

This is topological term in $\underline{d=1}$!

$\underline{d=1}$

$$A_L = iS\left(\frac{a}{S}\right)\int d\tau \int du \sum_x \Big\{ \vec{L}\cdot \partial_u \vec{N} \times \partial_\tau \vec{N} +$$
$$+ \vec{N}\cdot \partial_u \vec{L} \times \partial_\tau \vec{N} + \vec{N}\cdot \partial_u \vec{N} \times \partial_\tau \vec{L} \Big\}$$

Now, $\vec{N}\cdot\vec{L} = 0$

Also, $\partial_u \vec{N}\cdot\vec{N} = 0$ and $\partial_\tau \vec{N}\cdot\vec{N} = 0$

$\Rightarrow$ $\vec{L}$, $\partial_\tau \vec{N}$ and $\partial_u \vec{N}$ live in the same plane

$\Rightarrow$ $\vec{L}\cdot \partial_u \vec{N} \times \partial_\tau \vec{N} = 0$.

Also, $\sum_x a\{\ \} = \int dx \{\ \}$.

$$A_L = i\int d\tau \int dx \int_0^1 du \ \partial_u\left(\vec{N}\cdot\vec{L}\times\partial_\tau\vec{N}\right) - \cancel{\partial_u\vec{N}\cdot\vec{L}\times\partial_\tau\vec{N}}^{\,0}$$
$$- \underline{\vec{N}\cdot\vec{L}\times\partial_u\partial_\tau\vec{N}} + \partial_\tau\left(\vec{N}\cdot\partial_u\vec{N}\times\vec{L}\right) -$$
$$- \underline{\vec{N}\cdot\partial_u\partial_\tau\vec{N}\times\vec{L}} =$$

add up to $\emptyset$.

$$= i\int dx \int_0^1 du \underbrace{\int d\tau\ \partial_\tau\left(\vec{N}\cdot\partial_u\vec{N}\times\vec{L}\right)}_{\emptyset \text{ because of periodic BC}} +$$
$$+ i\int_0^\beta d\tau \int dx \underbrace{\int_0^1 du\ \partial_u\left(\vec{N}\cdot\vec{L}\times\partial_\tau\vec{N}\right)}_{\vec{N}(x,\tau,u=1)\cdot\vec{L}(x,\tau,u=1)\times\partial_\tau\vec{N}(x,\tau,u=1)}$$

$$= i\int dx \int_0^\beta d\tau\ \vec{N}(x,\tau)\cdot\vec{L}(x,\tau)\times\partial_\tau\vec{N}(x,\tau)$$

$$= -i\int dx \int d\tau\ \vec{L}\cdot\vec{N}\times\partial_\tau\vec{N}$$

$$\hat{H}_{AFM}(\vec{S}) = J \sum_{x} \vec{S}_{x} \cdot \vec{S}_{x+1} \rightarrow JS^2 \sum_{x} \vec{n}(x,\tau) \cdot \vec{n}(n+1,\tau) \quad H_{AFM}(\vec{n})$$

$$= \frac{JS^2}{2} \sum_{x} \left[\left(\vec{n}(x,\tau) + \vec{n}(x+1,\tau) \right)^2 - \underbrace{\vec{n}^2(x,\tau)}_{1} - \underbrace{\vec{n}^2(x+1,\tau)}_{1} \right]$$

$$= \frac{JS^2}{2} \sum_{x} \left[\vec{N}(x,\tau) - \vec{N}(x+1,\tau) + \frac{a}{S} \left(\vec{L}(x,\tau) + \vec{L}(x+1,\tau) \right) \right]^2$$

(+const)

$$= \frac{JS^2}{2} \sum_{x} \left(a \partial_x \vec{N}(x,\tau) \right)^2 + \frac{4a^2}{S^2} \vec{L}^2(x,\tau) =$$

$$= \frac{1}{2} \int dx \left\{ JS^2 a \left(\nabla_x \vec{N} \right)^2 + 4Ja \vec{L}^2 \right\}$$

$$Z = \int \mathcal{D}\vec{N}\,\mathcal{D}\vec{L}\ \exp - S_{AFM}$$

$$S_{AFM} = A_N + A_L + \int_0^\beta d\tau\, H_{AFM} =$$

$$= i\frac{\theta}{4\pi}\int dx \int d\tau\ \vec{N}\cdot\partial_x\vec{N}\times\partial_\tau\vec{N} + \frac{1}{2}\int dx\int d\tau\Big\{-2i\,\vec{L}\cdot\vec{N}\times\partial_\tau\vec{N}$$

$$+ JS^2a\,(\partial_x\vec{N})^2 + 4Ja\,\vec{L}^2\Big\}$$

Some manipulations are in order:

$$L_N = JS^2a\,(\partial_x\vec{N})^2 + 4Ja\left(\vec{L}^2 - \frac{i}{2Ja}\,\vec{L}\cdot\vec{N}\times\partial_\tau\vec{N}\right) =$$

$$= JS^2a\,(\partial_x\vec{N})^2 + 4Ja\left[\left(\vec{L} - \frac{i}{4Ja}\,\vec{N}\times\partial_\tau\vec{N}\right)^2 + \frac{1}{(4Ja)^2}\left(\vec{N}\times\partial_\tau\vec{N}\right)^2\right]$$

$$= 4Ja\left(\vec{L} - \frac{i}{4Ja}\,\vec{N}\times\partial_\tau\vec{N}\right)^2 + \frac{1}{4Ja}\left(\vec{N}\times\partial_\tau\vec{N}\right)^2$$

$$+ JS^2a\,(\partial_x\vec{N})^2$$

Now, $\left(\vec{N}\times\partial_\tau\vec{N}\right)^2 = \underbrace{\vec{N}\times\partial_\tau\vec{N}}_{A}\cdot\left(\underset{B}{\vec{N}}\times\underset{C}{\partial_\tau\vec{N}}\right) =$

$$\bar{A}\cdot\bar{B}\times\bar{C} = \bar{B}\cdot\bar{C}\times\bar{A}$$

$$= \vec{N}\cdot\left(\partial_\tau\vec{N}\times\vec{N}\times\partial_\tau\vec{N}\right) = \vec{N}\cdot\left(\vec{N}\,(\partial_\tau\vec{N})^2 - \partial_\tau\vec{N}\,(\partial_\tau\vec{N}\times\vec{N})\right)$$

$$\bar{a}\times\bar{b}\times\bar{c} = \bar{b}\,(\bar{a}\cdot\bar{c}) - \bar{c}\,(\bar{a}\cdot\bar{b})$$

$$= \underset{\substack{\parallel\\1}}{\vec{N}^2}\,(\partial_\tau\vec{N})^2 - \underbrace{\vec{N}\cdot\partial_\tau\vec{N}}_{\phi}\,(\partial_\tau\vec{N}\times\vec{N}) = (\partial_\tau\vec{N})^2$$

Thus obtain $L_N = 4Ja\left(\vec{L}'\right)^2 + \frac{1}{4Ja}\left(\partial_\tau \vec{N}\right)^2 + JS^2a\left(\partial_x \vec{N}\right)^2$

$\vec{L}' = \vec{L} - \frac{i}{4Ja}\vec{N}\times\partial_\tau\vec{N}$ "decouples" from the rest and can be integrated over:

$$Z = \int \mathcal{D}\vec{N} \exp - A_N - \frac{1}{2}\int dx\,d\tau\left[\frac{1}{4Ja}\left(\partial_\tau\vec{N}\right)^2 + JS^2a\left(\partial_x\vec{N}\right)^2\right]$$

$$\cdot \int \mathcal{D}\vec{L}' \exp - \frac{1}{2}\int dx\,d\tau\,(4Ja)\left(\vec{L}'\right)^2$$

$$= Z_N \cdot Z_L$$

$$Z_N = \int \mathcal{D}\vec{N} \exp - A_N - \frac{1}{2}\int dx\int d\tau\left[\frac{1}{4Ja}\left(\partial_\tau\vec{N}\right)^2 + JS^2a\left(\partial_x\vec{N}\right)^2\right]$$

We denote $S_N \equiv \frac{1}{2}\int dx\,d\tau\left[\frac{1}{4Ja}\left(\partial_\tau N\right)^2 + JS^2a\left(\partial_x\vec{N}\right)^2\right]$

Notice that τ-derivative and x-derivative appear <u>symmetrically</u>!

Re-write it as

$$S_N = \frac{1}{2gc}\int dx\int d\tau\left[\left(\partial_\tau\vec{N}\right)^2 + c^2\left(\partial_x\vec{N}\right)^2\right]$$

where $\frac{c^2}{g} = JS^2a$, $\frac{1}{gc} = \frac{1}{4Ja}$

Thus $\frac{1}{g^2} = \frac{JS^2 a}{4Ja} = \frac{S^2}{4} \Rightarrow g = \frac{2}{S}$ = <u>coupling constant</u>

$$c^2 = JS^2 a \cdot 4Ja = 4J^2 S^2 a^2$$

$$c = 2JSa$$ = <u>spin velocity</u>

(velocity of spin excitations)

$$S_N = \frac{1}{2gc} \int_0^\beta d\tau \int dx \left[(\partial_\tau \vec{N})^2 + c^2 (\partial_x \vec{N})^2 \right]$$

let $\tau = \frac{y}{c} \Rightarrow \partial_\tau \vec{N} = \partial_y \vec{N} \cdot \frac{\partial y}{\partial \tau} = c\, \partial_y \vec{N}$

Then $S_N = \frac{1}{2gc} \int_0^{c\beta} dy \frac{1}{c} \int dx \; c^2 (\partial_y \vec{N})^2 + c^2 (\partial_x \vec{N})^2 =$

$$= \frac{1}{2g} \int dx\, dy \left[(\partial_x \vec{N})^2 + (\partial_y \vec{N})^2 \right]$$

$$\vec{N} = (\cos\varphi \sin\theta,\ \sin\varphi \sin\theta,\ \cos\theta)$$

$$(\partial_x \vec{N})^2 = (\partial_x \varphi \ \sin\varphi \sin\theta)^2 + (\partial_x \theta \ \cos\varphi \cos\theta)^2$$

$$+ (\partial_x \varphi \cos\varphi \sin\theta)^2 + (\partial_x \theta \ \sin\varphi \cos\theta)^2 + (\partial_x \theta \sin\theta)^2$$

$$= (\partial_x \varphi)^2 (\sin^2\varphi \ \sin^2\theta + \cos^2\varphi \ \sin^2\theta)$$

$$+ (\partial_x \theta)^2 \ (\cos^2\varphi \cos^2\theta + \sin^2\varphi \cos^2\theta + \sin^2\theta)$$

$$= (\partial_x \theta)^2 + \sin^2\theta \ (\partial_x \varphi)^2$$

Clearly

$$(\partial_y \vec{N})^2 = (\partial_y \theta)^2 + \sin^2\theta (\partial_y \varphi)^2$$

2d theory

$$S_N = \frac{1}{2g} \int dx dy \left\{ (\vec{\nabla} \theta)^2 + \sin^2\theta (\vec{\nabla} \varphi)^2 \right\}$$

where $\vec{\nabla} \theta = \hat{x} \ \partial_x \theta + \hat{y} \ \partial_y \theta$.

Topological meaning

$\theta = 2\pi S$ The meaning of $A_N = \frac{\theta}{4\pi} \int dx\, d\tau \left(\vec{N} \cdot \partial_x \vec{N} \times \partial_\tau \vec{N} \right)$

$$= \frac{\theta}{4\pi} \int dx \frac{dy}{c} \left(\vec{N} \cdot \partial_x \vec{N} \times c\, \partial_y \vec{N} \right)$$

$$= \frac{\theta}{4\pi} \int dx\, dy \left(\vec{N} \cdot \partial_x \vec{N} \times \partial_y \vec{N} \right)$$

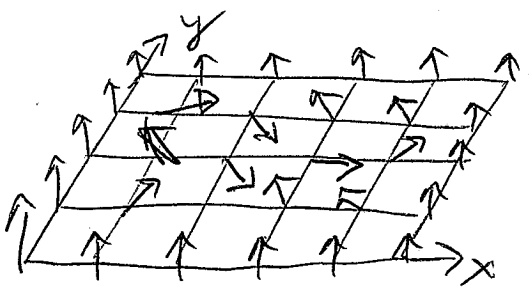
$$= \frac{\theta}{8\pi} \int dx\, dy \left(\varepsilon_{\mu\nu} \vec{N} \cdot \partial_\mu \vec{N} \times \partial_\nu \vec{N} \right) = \text{area of the } \vec{N} \text{ sphere traced by } \vec{N}.$$

$$\varepsilon_{xy} = +1 \quad , \varepsilon_{yx} = -1 \quad , \varepsilon_{xx} = \varepsilon_{yy} = 0$$

Pontryagin index (winding number) of the spin configuration $\{\vec{N}(x,y)\}$

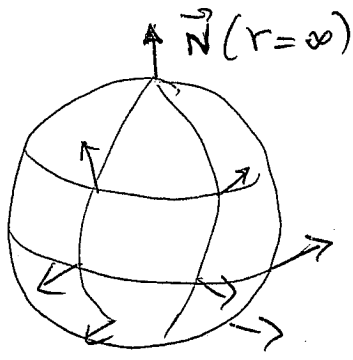
$$Q = \frac{1}{8\pi} \int dx\, dy \left(\varepsilon_{\mu\nu} \vec{N} \cdot \partial_\mu \vec{N} \times \partial_\nu \vec{N} \right)$$

physical space



fix $\vec{N}(\vec{r})$ at $|\vec{r}| = \infty$ so that S_N is finite ($S_N \sim$ volume $\cdot (\partial \vec{N})^2 = 0$ if $\underline{\vec{N} = \text{const}}$)

order parameter space



Every $\vec{N}(x,y)$ maps onto sphere S_2 representing order parameter space.

sphere S_2: $|\vec{N}|^2 = 1$

Q counts how many times $\{\vec{N}\}$ has wrapped around the sphere